Global Metals & Mining

EV battery metal recycling: The future of battery metals

While today nearly all EV-battery related metals come from mined reserves, we estimate recycled metals from retired EV batteries could supply nearly half of the EV market by 2040, and potentially 70-80% in the long run. We expect the TAM for EV battery metal recycling to reach over US$200bn/year globally and over US$60bn/year in China in the long run, versus only US$10mn currently.

We view EV battery metal recycling as indispensable for sustainable EV adoption, by addressing the long term supply constraints on resources and reducing dependence on mined metals. The global EV market could grow by 13x by 2040E or over 25x if EV penetration reaches 100%. Recycling with EV battery metals could extend the global upstream reserve by 25-70 years, all else equal. In addition, the use of recycled materials may also cut CO_2 emissions by 7-17% during battery manufacturing, under the right technologies.

There are uncertainties and challenges - gaps need to be addressed in regulation and recycling technologies before the market scales up. Stricter regulation will be needed to ensure full collection and recycling by professionals. More standardized EV batteries will help develop more efficient technology for the pre-treatment stage. More importantly, we believe competition will rise: OEMs and battery manufacturers are gaining more bargaining power over raw material resources by owning the majority of retired EV batteries and scrap.

We initiate GEM (Buy, 12-m target price Rmb16.0) and Huayou Cobalt (Buy, 12-m target price Rmb128.0) as industry leaders in EV battery metal recycling in China positioned to benefit from the growing recycling market. We also upgrade earnings and TPs for Ganfeng-H/A (12-m target prices to HK$246/Rmb256; up ~7%) to incorporate its growing lithium recycling business.

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EV Battery Metal Recycling in numbers

A mini EV Wuling Hongguang consumes **55kg** lithium while an iPhone only consumes **11-13g** lithium and cobalt.

Currently, battery related demand accounts for **75%** of global lithium demand, **60%** of global cobalt demand and **10%** of global nickel demand.

Global battery demand for lithium and nickel will be **12-13x** of the current size, **2x** of the current size for cobalt by 2040E.

An EV battery is usually retired after **6-8** years of usage, when capacity falls below **80%** of the initial level.

By 2030, EV batteries coming to **retirement stage** will be the same size as the current annual battery production size at **300-500GWh.**

Metal recovery rates have reached over **95%** for nickel and cobalt, above **90%** for lithium currently, and recycled metals can be reused in EV battery manufacturing.

Recycled metals could supply **39-57%** of battery demand by 2040E, and potentially **70-80%** in the long run.

If no recycling, mine reserve life will be only **17-30 years** for lithium, cobalt and nickel under long term battery demand. Recycling with EV battery materials could extend the global upstream reserve by **25-70 years**, all else equal.

TAM for EV battery material recycling could reach over **US$200bn/year** globally and over **US$60bn/year** in China in the long run.

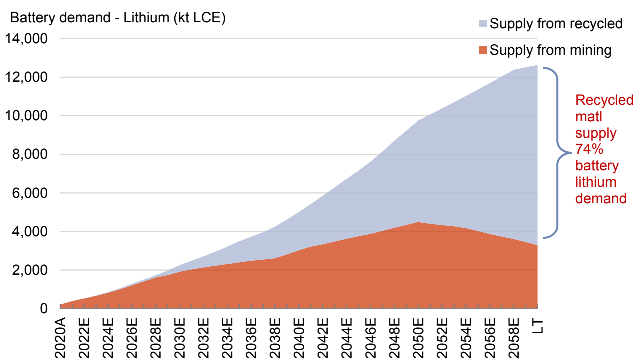
EV battery materials recycling can reduce CO_2 emission by **7-17%** during battery manufacturing under right technologies.

Currently, most retired battery dismantling is done **manually** due to **inconsistent** standard of battery systems.

Source: Goldman Sachs Global Investment Research

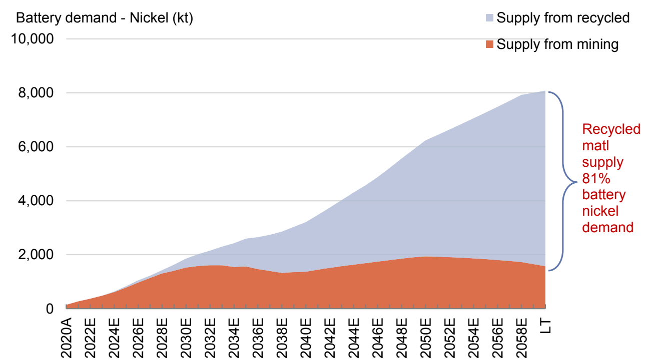
Key charts

Exhibit 1: Mined lithium demand from battery - with and without recycling



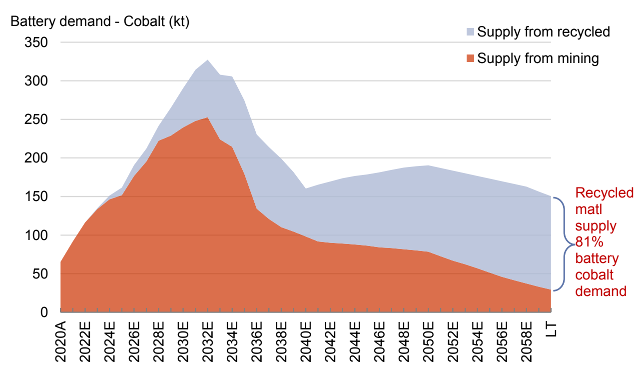
Source: SMM, Goldman Sachs Global Investment Research

Exhibit 2: Mined nickel demand from battery - with and without recycling



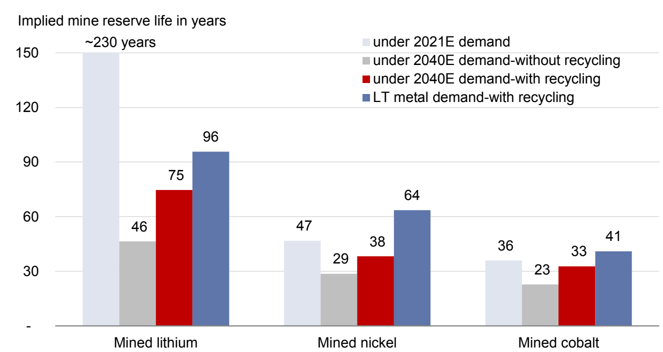
Source: SMM, Goldman Sachs Global Investment Research

Exhibit 3: Mined cobalt demand from battery - with and without recycling



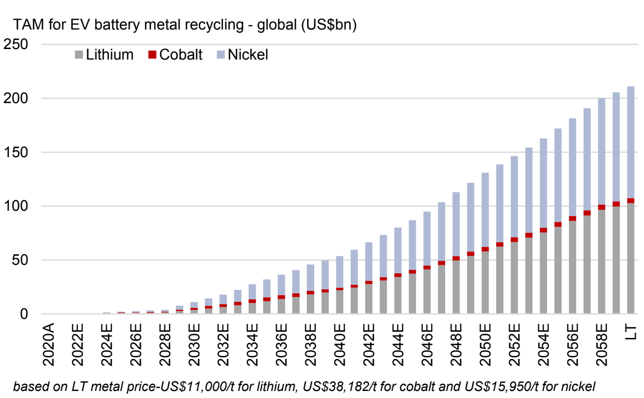
Source: SMM, Goldman Sachs Global Investment Research

Exhibit 4: Implied mine reserve life - with and without recycling



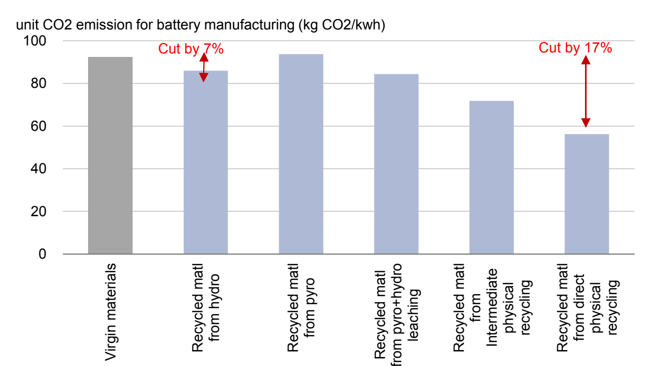
Source: USGS, Goldman Sachs Global Investment Research

Exhibit 5: TAM for EV battery metal recycling - global



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 6: Unit CO2 emission for battery manufacturing by using virgin material versus recycling matl - CO2 emission can cut by 7-17% in most cases for recycling



Romare is a study by IVL Swedish Environmental Research Institute, authored by Mia Romare, published in 2017

Source: Romare (2017): The Life Cycle Energy Consumption and Greenhouse Gas Emissions from Lithium-Ion Batteries, ICCT, Goldman Sachs Global Investment Research

EV battery metal recycling - questions that may be on top of mind

How important is EV battery metal recycling?

With strong EV growth in next 1-2 decades, battery demand from major minor metals including lithium, cobalt and nickel will grow more than 10x the current size. There're rising concerns on sustainable metal supply as one of key bottlenecks for EV sustainable development. As EV batteries usually retire after 6-8 years of usage but still contain many valuable key metals, recycling of retired EV battery metals is an indispensable part for sustainable EV adoption by addressing the long term supply constraint on resources and reducing the dependence on mined metals. We estimate the full recycling of EV battery metals could supply 39-57% of lithium, cobalt and nickel demand for batteries by 2040E or 70-80% once EV penetration reaches 100% for 8-10 years, partly realizing self-circulation within EV battery chain. Mine reserve life for these minor metals could extend by 100%-240% for lithium, cobalt and nickel with deceleration of mined metals demand compared to no recycling.

How should investor think about recycled EV battery metal market in the long run?

Based on our long term metal prices forecasts, we expect TAM for EV battery metal recycling could grow over US$200bn/year globally and over US$60bn/year in China in the long term, implying strong growth potential along with growing EV market. Currently, as most EVs hasn't come to retirement stage, the major source for EV battery recycling is battery scrap or recalled batteries. Currently global market is only at US$10mn under our estimates.

How much greener?

EV battery recycling is green enabler as it reduces CO2 emission by 7-17% during battery manufacturing in most cases. On LCA (Life-cycle assessment) basis, we estimate EV battery recycling could further save CO2 emissions for EV by another 1-2%. Also the early retirement (8 years) of EV battery will not necessarily lead to higher CO2 emission for EV compared to ICE car derived from ICCT recent study on CO2 emission on LCA basis for EV and ICE car.

What are the major challenges for EV battery recycling?

Mostly in regulation and technologies. Stricter regulations are required to ensure the proper retirement, regulated collection and recycling treatment to fully utilise the retired batteries. On the other hand, EV battery recycling technology is still at early stages especially for pre-treatment processes including testing, dismantling, discharging and separation. Standardized EV battery is needed to improve the efficiency of pre-treatment under rising recycling volume. Further technology development in metal extraction are mostly on improving recovery rate, shortening the recycling process and lowering the cost.

What are they key technologies in recycling process?

EV batteries typically need to retire from EVs when the capacity falls below 80% of the initial level. Batteries with 40-80% of their initial capacity could potentially be repurposed for second-life application such as energy storage and low-speed vehicles. Batteries that do not meet the minimum requirements for second-life reuse can be recycled. In the recycle process, retired batteries need to be pre-treated first, including discharging, dismantling, crushing, and physical separation. Fine powder called "black mass" is produced after the pretreatment. Currently, the major processes for recycling treatment that are operating at industrial scale includes pyrometallurgical and hydrometallurgical. Overall, second-life reuse has higher requirements for retired EV battery in terms of consistency in capacity, size, inner structure, economics versus new battery.

EV battery recycling ecosystem

Exhibit 7: EV battery recycling ecosystem

EV battery recycling ecosystem					
Stages	**Collection**	**Transportation**	**Second-life reuse**	**Recycling**	**Compounds producer**
Key participants	• EV OEMs • Battery manufacturers • Battery material producers • Battery renting companies • EV fleet operators • 3rd party recycling service providers • Traders • Unlicensed family workshops • MIIT whitelist companies	• 3rd party service providers • Traders • EV OEMs • Battery manufacturers • Transportation and Logistics company • MIIT whitelist companies	• Specialized second-life battery producers/repurposing companies • Battery producers • EV OEMs • Grid operators • Energy storage system operators • MIIT whitelist companies	• Specialized recycling companies • Unlicensed family workshops • Battery materials producer • EV OEMs • Battery producer • 3rd party service provider	• Lithium compounds producers • Battery materials producers • Chemicals company
Process & Technology	**Main collection channels:** • Direct sale to MIIT whitelist recycling companies by OEMs, battery producers, and fleet operator • Sales to recycling service provider and traders, who could potentially re-sell to recycling company or unlicensed mills	**Transportation requirement:** • Special transportation for hazardous materials under regulation	**Second-life reuse process:** • Dismantling battery pack • Battery pack assessment • Characterization test for modules and cells • Rearranging cells into modules for second-life battery **Key equipment:** • Battery testing equipment **Major applications:** • Energy storage systems (ESS) • Low-speed E-mobile	**Pretreatment** • Deactivation & Discharging • Dismantling • Crushing • Physical separation **Recycling technology:** • Pyrometallurgical • Hydrometallurgical • Direct process **Key equipment:** • Shredder • Sieving machine • Magnetic separator • Rotary Kiln	**Main end products:** • Lithium carbonate • Ternary precursor • Ternary cathode materials • Cobalt oxide • LCO
Revenue, Cost & Return	**Battery acquisition cost:** Rmb24,300/t for NMC523	**Transportation cost for hazardous materials:** Rmb10-12/t.km	**Second-life LFP battery selling price:** Rmb0.64-0.68/Wh **Producer margin:** 15-20%	**3rd party service provide: Recycling revenue:** Rmb13,000-46,000/t **Recycling cost:** Rmb13,000-35,000/t **EBITDA/Capex:** 1%-19%	**Battery matl producer: Revenue from recycled matl:** Rmb170,000/t (end product) **Cost:** Rmb120,000-150,000/t **EBITDA/Capex:** 19%-89%
Major players	**China:** • Huayou (603799.SS) • GEM (002340.SZ) • Ganfeng (1772.HK/002460.SZ) • Ganzhou Highpower (non-listed) • Guangdong Guanghua (002741.SZ) • BAIC (1958.HK) • Tianjin Yinlong (603969.SH) • BYD (1211.HK/002594.SZ) • CATL/Bangpu (300750.SZ) • Ruicycle (non-listed) • Zhongwei (300919.SZ) • Xiamen Tungsten (600549.SS) • Didi (DIDI) **Global:** • Tesla (TSLA) • Volkswagen (VOW3.DE) • Nissan (7201.T) • Panasonic (6752.T) • Redwood Materials (non-listed) • Accurec Recycling (non-listed) • Redux (non-listed) • Neometals (NMT.AX) • Umicore (UMI.BR) • Li-Cycle Corp (LICY)	**China:** • Huayou (603799.SS) • GEM (002340.SZ) • Ganfeng (1772.HK/002460.SZ) • Ganzhou Highpower (non-listed) • CATL/Bangpu (300750.SZ) • Ruicycle (non-listed) • Zhongwei (300919.SZ) • Xiamen Tungsten (600549.SS) **Global:** • Panasonic (6752.T) • Redwood Materials (non-listed) • Accurec Recycling (non-listed) • Redux (non-listed) • Neometals (NMT.AX) • Umicore (UMI.BR) • Li-Cycle Corp (LICY)	**China:** • GEM (002340.SZ) • Ganzhou Highpower (non-listed) • Guangdong Guanghua (002741.SZ) • BAIC (1958.HK) • Tianjin Yinlong (603969.SH) • BYD (1211.HK/002594.SZ) • CATL/Bangpu (300750.SZ) • Zhejiang Tianneng (non-listed) • Anhui Lvwo (non-listed) • Zhongtian Hongliqingyuan (non-listed) • Henan Liwei New Materials (non-listed) • Qiantai Technology (non-listed) • Zhuhai Zhongli New Materials (non-listed) • Ruicycle (non-listed) • Chonghonggerun (non-listed) **Global:** • Tesla (TSLA) • Volkswagen (VOW3.DE) • Nissan (7201.T) **Downstream:** • China Tower (0788.HK) • State Grid (non-listed) • ENGIE (ENGI.FR)	**China:** • Huayou (603799.SS) • GEM (002340.SZ) • Ganfeng (1772.HK/002460.SZ) • Ganzhou Highpower (non-listed) • Guangdong Guanghua (002741.SZ) • BYD (1211.HK/002594.SZ) • CATL/Bangpu (300750.SZ) • Jinyuan New Materials (871370) • Ruicycle (non-listed) • Jiana (non-listed) • Zhongwei (300919.SZ) • Xiamen Tungsten (600549.SS) **Global:** • Tesla (TSLA) • Volkswagen (VOW3.DE) • Nissan (7201.T) • Redwood Materials (non-listed) • Reverse Logistics Group (non-listed) • Accurec Recycling (non-listed) • Redux (non-listed) • Neometals (NMT.AX) • Umicore (UMI.BR)	**China** • Huayou (603799.SS) • GEM (002340.SZ) • Ganfeng (1772.HK/002460.SZ) • Zhongwei (300919.SZ) • Jiana (non-listed) • Xiamen Tungsten (600549.SS) **Global:** • Neometals (NMT.AX) • Umicore (UMI.BR)

Source: Company data, Goldman Sachs Global Investment Research

EV battery metal recycling crucial for EV growth

With strong EV growth in 2021, key metals for battery making ran into certain shortages and surging prices- lithium carbonate/cobalt/nickel prices increased by 23-434% yoy and cost pressure increased for EV manufacturing. As EV growth will remain strong in the next 1-2 decades - 13x of current size by 2040E or over 25x if EV penetration reaches 100% under our global auto's team estimates, there're rising concerns on sustainable metal supply as one of key bottlenecks for EV development. Under this situation, we view EV battery metal recycling as indispensable for sustainable EV adoption by addressing the long term supply constraints on resources and reducing dependence on mined metals. As EV batteries usually retire after 6-8 years of usage but still contain many valuable key metals, we estimate the full recycling of EV battery metals could supply 39-57% of lithium, cobalt and nickel demand for batteries by 2040E or 70-80% once EV penetration reaches 100% for 8-10 years, partly realizing self-circulation within EV battery chain. Mine reserve life for these minor metals could extend by 100%-240% for lithium, cobalt and nickel with deceleration of mined metals demand compared to no recycling.

Based on our long term metal prices forecasts, we expect TAM for EV battery metal recycling could grow over US$200bn/year globally and over US$60bn/year in China in the long term, implying strong growth potential along with growing EV market.

Rising depletion risk for minor metals under strong EV growth

An EV car battery is made of several minor metals with the most critical ones being lithium, cobalt and nickel. These metals are currently used in the cathode materials of EV batteries. An EV car is usually equipped with a battery size ranging between 10-100KWh, based on the current most popular EV models including Wuling Hongguang MINI EV, BYD E2, Tesla Model 3 and NIO-ES6. Lithium is used in almost all types of batteries including LFP and ternary. Depending on the battery type used (LFP or NCM811, NCA), lithium usage per KWh battery is about 0.55kg LCE to 0.82kg LCE. Cobalt and nickel are only used in ternary battery cathode materials - cobalt usage per KWh battery is about 0.07-0.22kg for various types of ternary battery, and nickel usage per KWh is 0.67-0.87kg for various types of ternary battery (based on various company data). For a Tesla Model 3 using an NCA battery at 76.8KWh, for example, the metal usage would be 55kg lithium, 5kg cobalt and 66kg nickel, which is several thousands times higher than an iPhone 13.

Based on current global EV market size (over 6mn units), we estimate battery related demand is 390kt LCE for lithium (75% of total lithium demand), 92kt for cobalt (60% of total cobalt demand) and 266kt (10% of total nickel demand) in 2021E. With the rapid growth of EVs, battery demand from these metals will grow to more than 10x the current size - we estimate that by 2040E, global battery demand from lithium and nickel will grow by 12-13x of the current size, to reach 5.0mnt LCE and 3.2mnt, while cobalt will be nearly 2x the current size at 160kt despite the trend of cutting usage of cobalt in batteries. Based on 2040E annual demand size, we estimate the implied mine reserve life will be only 46 years for lithium, 29 years for cobalt and 23 years for nickel, even considering reserve increases through continuous exploration. This is much lower than implied reserve life using current demand size at 229 years for lithium, 47 years for cobalt and 36 years for nickel, suggesting a rising depletion risk for critical minor metals.

Exhibit 8: Minor metal usage for major EV and mobile phone models

User type	Battery needed in KWh	Key metal usage
NIO-ES6	NCM811 100KWh	Li: 73kg Co: 11kg Ni: 87kg
Tesla Model 3	NCA 76.8KWh	Li: 55kg Co: 5kg Ni: 66kg
Wuling Hongguang Mini EV	LFP 13.8KWh	Li: 55kg
Apple Iphone 13	LCO 0.0134KWh	Li: 0.011kg Co: 0.013kg

Only includes cathode materials usage

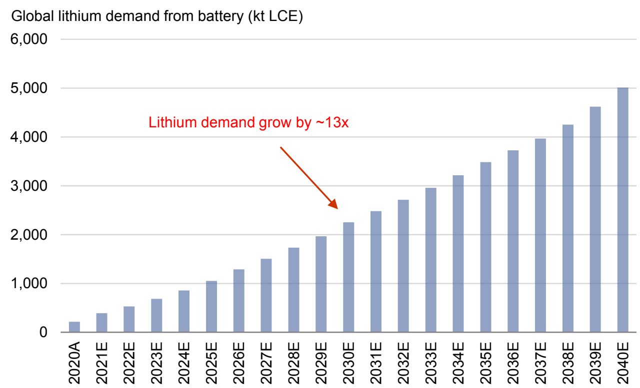
Source: Company data, Goldman Sachs Global Investment Research

Metal recovered from recycling can support majority of battery demand

With the EV acceleration and limited mine supply of critical minor metals as suggested above, EV battery metal recycling will be crucial for EV growth and sustainable EV adoption in the long run, in our view. An EV battery is usually retired after 6-8 years of usage (based on data from several industry sources), when capacity falls below 80% of the initial level, but still contains many key metals that can be reused. Therefore, we believe retired EV battery material recycling will gradually become another important source for critical metal supply including lithium, nickel and cobalt as a part of "Urban Mining". The "Urban Mining" concept started very early in the 1980s and refers to the recycling of electronic waste in cities via the refining process. However, we believe the urgency is much stronger than before post-EV development, with a much higher critical metal content per unit of vehicle compared to electronics products. For example, an iPhone 13 only features a 13.4Wh battery compared to the smallest size EVs, such as the Hongguang Mini, which uses a 1,000x larger 13.8KWh battery. As such, EV growth will lead to much greater usage of critical metals like lithium, cobalt and nickel, making recycling even more necessary.

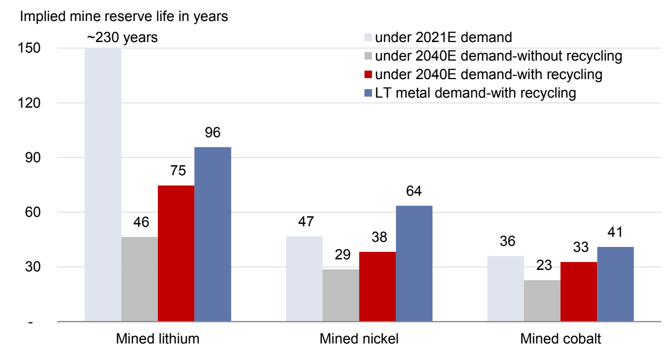
Under the assumption of an 8-year EV battery life before going to retirement stage and full collection and recycling materials, we estimate metals recovered from recycled EV batteries will be able to supply 40% of lithium, 39% of cobalt, and 57% of nickel demand for use in EV batteries by 2040E. These percentages are calculated under our GS global auto team's annual EV forecast by 2040E. Furthermore, if EV penetration were to reach 100% for 8-10 years, we estimate metals recovered from EV battery recycling could potentially supply 70-80% of lithium, cobalt and nickel for battery usage, realizing self-circulation within the EV supply chain. With rising self-circulation under recycling, mine reserve life could be extended with less usage of mined metal - we expect implied mine reserve life to increase by 34%-61% to 75 years for lithium, 38 years for cobalt and 33 years for nickel with EV battery recycling by 2040E or extend reserve by 25-70 years in the long run comparing to no recycling. This is due to deceleration of metal demand from mining with a rising volume of recovered metal supply - we estimate mined lithium demand growth could decelerate to a 5% CAGR during 2030-2040, and mined nickel/cobalt demand to decline by a 1-9% CAGR during 2030-2040.

Exhibit 9: Global battery demand from lithium to grow by 13x by 2040E



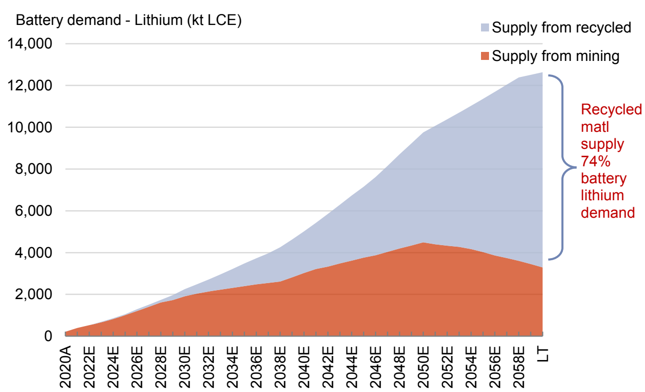
Source: SMM, Goldman Sachs Global Investment Research

Exhibit 10: Implied mine reserve life - with and without recycling



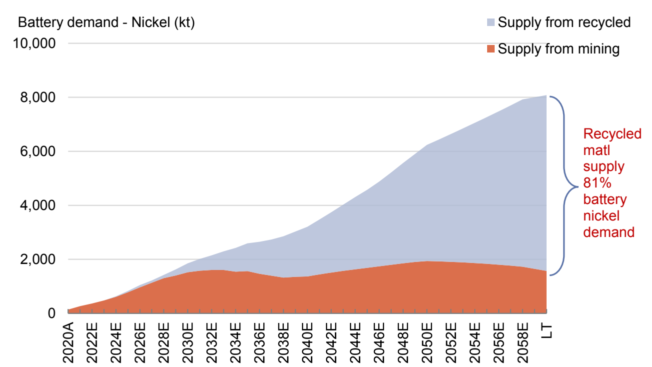
Source: USGS, Goldman Sachs Global Investment Research

Exhibit 11: Mined lithium demand from battery - with and without recycling



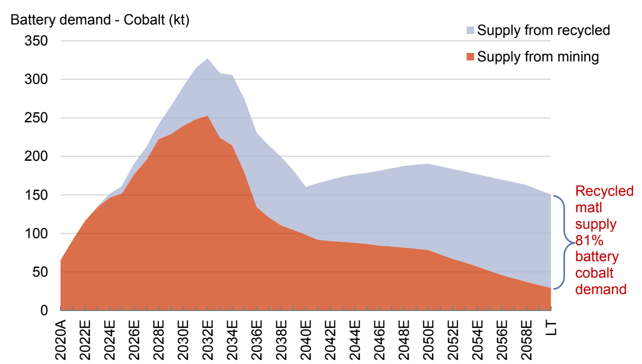
Source: SMM, Goldman Sachs Global Investment Research

Exhibit 12: Mined nickel demand from battery - with and without recycling



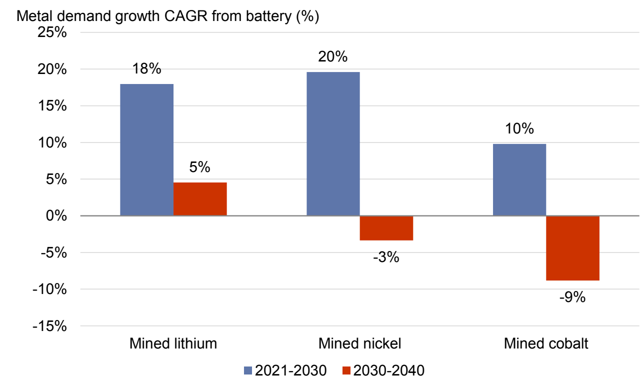
Source: SMM, Goldman Sachs Global Investment Research

Exhibit 13: Mined cobalt demand from battery - with and without recycling



Source: SMM, Goldman Sachs Global Investment Research

Exhibit 14: Mined metals demand growth should decelerate if recycling accelerates



Source: Goldman Sachs Global Investment Research

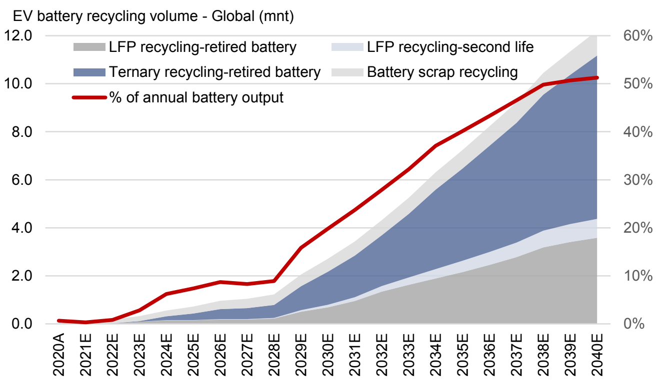
TAM for EV battery metal recycling market

Considering the future trend of EV acceleration, and the valuable critical materials left post-retirement of an EV battery, EV supply chain participants and 3rd party service providers are increasing capex to build infrastructure for retired EV battery collection and optimizing technologies to improve recovery rates for metals at lower treatment cost, aiming to develop an alternative source for raw material supply. With years of technology development and refining experience accumulated through treating electronic waste, recovery rates for key metals in EV battery recycling have already reached over 90% for lithium, with nickel and cobalt at over 95%. Currently, as most EVs have not reached the retirement stage, the major source for EV battery recycling is battery scrap or recalled batteries. According to Li-Cycle, 5-10% of battery production is typically rejected as waste during the manufacturing process and the percentage of scrap is even higher when battery plants are in the ramp-up stage.

Based on our GS global auto team's EV forecast of 81mn new car sales (PHEV+BEV) and related battery demand of 5,508GWh by 2040E, we expect annual battery demand to reach 6,600GWh including additional 1,000GWh replacement demand. Assuming an 8-year EV battery life before retirement and 5% of battery scrap generation during battery production, we expect total retired batteries and battery scrap available for recycling to increase to 3,719GWh by 2040E globally, over 100x the current size (at less than 40GWh). This would translate to 12mnt of retired EV batteries and scrap available for recycling, among which we estimate ternary batteries will account for over 60%. For China, we expect total retired batteries and battery scrap available for recycling to increase to 1,350GWh by 2040E, 1/3 of the global total and more than 100x the current size of 12GWh. Accordingly, this would translate to 5.7mnt retired EV batteries/scrap available for recycling, with LFP batteries accounting for around 70% of the total by 2040E. Interestingly, battery scrap accounts for the major part of EV battery recycling at the early stage (above 50%) but should gradually come down to a single digit due to a slower rate of growth than the pace of EV battery retirement. Notably, we estimate that China's EV batteries available for recycling will account for over 50% of the global total by 2030E due to the earlier EV development than in other countries, and will gradually decrease to 33% by 2040E.

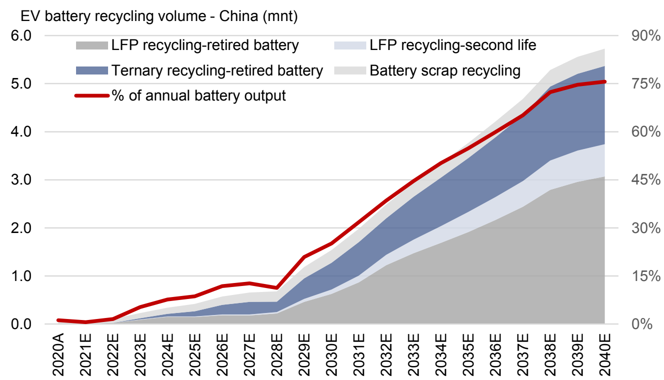
Based on key metals including lithium, cobalt and nickel recovered from EV battery metal recycling, and our annual price forecasts for these metals including long-term forecasts of US$11,000/t for lithium, US$38,182/t for cobalt and US$15,950/t for nickel, we expect the TAM for global EV battery metal recycling to reach US$54bn in 2040E, or over US$200bn per year for the long term, and US$15bn in 2040E, or over US$60bn per year, in China.

Exhibit 15: EV batteries and scrap available for recycling - Global



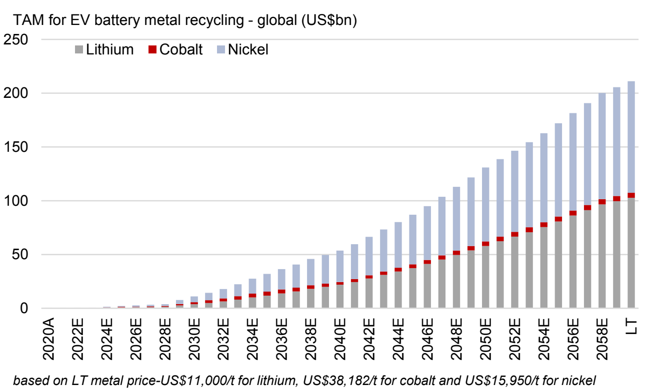
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 16: EV batteries and scrap available for recycling - China



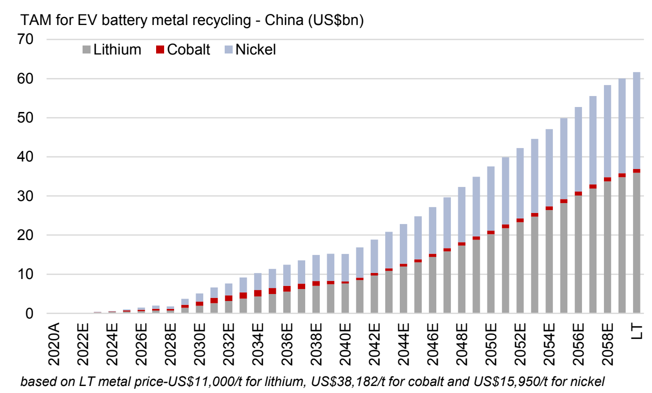
Source: SMM, Goldman Sachs Global Investment Research

Exhibit 17: TAM for EV battery metal recycling - global



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 18: TAM for EV battery metal recycling - China



Source: SMM, Goldman Sachs Global Investment Research

Exhibit 19: EV battery metal recycling model and EV-related metal demand from battery

		2020A	2021E	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2035E	2040E	LT
Global EV battery output - total	GWh	186	377	546	723	938	1,193	1,504	1,783	2,077	2,380	2,753	4,447	6,609	20,949
Global EV battery output - new car	GWh	155	314	455	603	782	994	1,254	1,486	1,731	1,983	2,294	3,706	5,508	10,891
Global EV battery output - replacement	GWh	31	63	91	121	156	199	251	297	346	397	459	741	1,102	10,058
Global EV battery to retire - annual	GWh	1	1	4	20	58	88	131	148	185	377	546	1,785	3,388	17,212
Retired as % of annual EV battery output	*%*	*1%*	*0%*	*1%*	*3%*	*6%*	*7%*	*9%*	*8%*	*9%*	*16%*	*20%*	*40%*	*51%*	*82%*
Global EV battery scrap - annual	GWh	9	19	27	36	47	60	75	89	104	119	138	222	330	n.a.
Scrap as % of EV battery output	*%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*n.a.*
Global EV battery to retire - annual	kt	8	7	26	118	314	432	610	651	785	1,565	2,170	6,471	11,174	56,763
Global EV battery scrap - annual	kt	58	112	157	201	244	290	349	397	444	489	544	771	1,038	n.a.
Lithium demand from battery - annual	kt LCE	215	390	527	683	854	1,051	1,290	1,505	1,735	1,967	2,253	3,484	5,011	12,634
yoy or CAGR	*%*	*18%*	*82%*	*35%*	*29%*	*25%*	*23%*	*23%*	*17%*	*15%*	*13%*	*16%*	*22%*	*8%*	*5%*
Supply from mining	kt LCE	214	389	525	671	818	995	1,206	1,410	1,617	1,729	1,910	2,407	3,020	3,292
Supply from EV battery matl recycling	kt LCE	1	1	3	12	36	56	84	95	118	238	343	1,076	1,992	9,342
Cobalt demand from battery - annual	kt	65	92	116	135	151	162	191	212	242	265	291	275	160	150
yoy or CAGR	*%*	*-4%*	*40%*	*27%*	*16%*	*12%*	*7%*	*18%*	*11%*	*14%*	*10%*	*12%*	*8%*	*-6%*	*0%*
Supply from mining	kt	65	92	116	134	146	152	177	195	222	229	240	179	98	29
Supply from EV battery matl recycling	kt	-	-	-	1	5	10	14	17	19	36	51	95	62	121
Nickel demand from battery - annual	kt	141	266	368	481	635	828	1,047	1,222	1,421	1,629	1,857	2,593	3,206	8,082
yoy or CAGR	*%*	*31%*	*88%*	*38%*	*31%*	*32%*	*30%*	*26%*	*17%*	*16%*	*15%*	*18%*	*20%*	*6%*	*5%*
Supply from mining	kt	141	266	368	477	610	780	969	1,132	1,305	1,402	1,526	1,565	1,366	1,575
Supply from EV battery matl recycling	kt	-	-	-	3	25	48	78	90	115	227	331	1,028	1,840	6,508

Source: Goldman Sachs Global Investment Research

Why EV battery service life is shorter than claimed?

EVs sold in China or global usually have a minimum warranty period of 8 years for the battery, but the battery service life is shorter than that at only 4-6 years in some cases. An EV battery pack is made from a number of battery cells, and several factors in battery manufacturing could lead to deviation in voltage, capacity, and internal resistance even for the same battery pack. While the performance of a battery pack is determined by the performance of cells, it is by no means a simple accumulation of the performance of the single cells. Due to the inconsistency of cell performance, when a battery pack is repeatedly used in an EV, various problems lead to a shortened service life. Reported battery life data is mostly derived from lab/factory tests under certain conditions like temperature, humidity and pressure, and could be different in real-life settings and dependent on the EV driver's use habits. Several factors that might lead to shorter battery life include: 1) the trend of shifting to high-nickel ternary batteries, which have a full charge cycle of 800-1,000 times versus 2,000 times for an LFP battery, 2) fast charging that reduces battery life due to more heat generation could lead to decomposition of the battery structure, 3) use of air-cooling and entertainment systems, and 4) weather conditions, both hot and cold.

What could potentially increase the battery service life of solid-state lithium metal battery technology? According to Quantumscape, an all-solid-state-battery could prolong battery life by eliminating electrolytes in conventional lithium-ion cells. In addition, a Harvard research paper published on 12 May, 2021 in Nature "A dynamic stability design strategy for lithium metal solid state batteries" authored by Luhan Ye & Xin Li, claimed to have designed a lithium metal solid-state battery that can be fully charged at least 10,000 times at high density, which could prolong EV battery service life to 10-15 years without the need to replace the battery.

EV battery recycling is green enabler

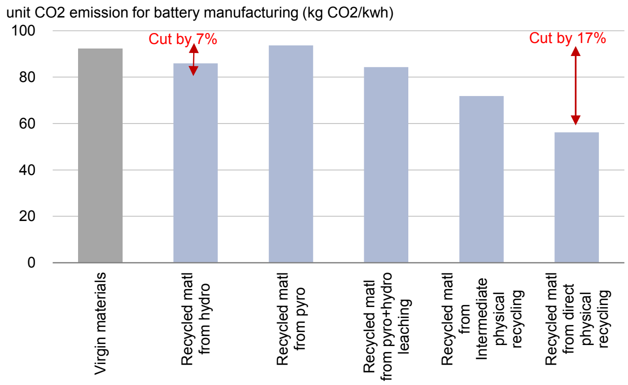
EV battery recycling is green enabler as it reduces CO_2 emission by 7-17% during battery manufacturing. On LCA (Life-cycle assessment) basis, we note EV has lower CO_2 emission than ICE car and EV battery recycling could further save CO_2 emissions for EV by another 1-2%. Besides, we found that early retirement (8 years) of EV battery will not necessarily lead to higher CO_2 emission for EV compared to ICE car derived from ICCT recent study on CO_2 emission on LCA basis for EV and ICE car.

EV battery recycling can save CO_2 emissions for battery manufacturing in most cases and this will depend on recycling technologies used. On the battery manufacturing emission perspective, battery recycling can save CO_2 emissions by 1-2.5kg CO_2e per kg of battery recycled according to a study by IVL Swedish Environmental Research Institute, this would translate to 7%-17% net reduction of CO_2 emission for battery manufacturing according to ICCT or 1-2% on EV LCA basis. Specifically, the Institute estimates that the direct CO_2 emission from a hydrometallurgical process is around 2.5kg/kg battery, with the major sources of emission being acid leaching and cell separation. Since recycled materials can substitute virgin materials, hydrometallurgical recycling process will generate a CO_2 credit of around 3.5kg/kg battery, thus leading to a net CO_2 savings of 1kg/kg battery, 7% of total CO_2 emission for battery manufacturing. In comparison, the more energy-intensive pyrometallurgical process could cause a net increase of 1.2kg CO_2/kg battery, or translating to an 8% net increase of CO_2 emissions. Direct physical recycling can deliver the most of CO_2 emission savings at 17% at 2.5kg/kg battery.

On EV LCA perspective, the early retirement of EV battery is not necessarily leading to higher CO_2 emission for EV compared to ICE car. According to ICCT recent study, battery production in different regions have different emission intensity with highest in China at 51-77kg/KWh, and lowest in Europe at 34-57kg/KWh depending on different cathode chemistry, translating to 7-15gCO_2/km on LCA basis if using life cycle mileage of 200,000 km, or 14-30g CO_2/km on LCA basis under life cycle mileage of 100,000km. The same study highlighted that a BEV car GHG emission is about 83-153g CO_2e/km for EU, US and China, 41%-67% lower than that for ICE. We estimate if life cycle mileage is shortened to eight years with early retirement of EV battery, GHG emission for EV is still 37-53g CO_2/km lower than ICE car in the US and EU even with additional CO_2 emission from replacement of one new battery made from recycled materials, but is higher than ICE car in China due to higher emission for EV. However, with continued improving power mix by 2030E and by adding a new battery made from recycled materials for cars registered in 2030E we estimate GHG emission for EV will be 25-96g CO_2/km lower than ICE car in the US, EU and China, based on ICCT report.

Based on the single car emission estimates for EVs and ICEs, and GS EV forecast by 2040E, we expect weighted average unit CO_2 emission on LCA basis to decline to 134g CO_2/km by 2040E if no early retirement or 185g CO_2/km if considering early retirement of battery at eight years service, or 195g CO_2/km if considering one battery replacement with recycled materials, all lower than ICE emission at 228g CO_2/km.

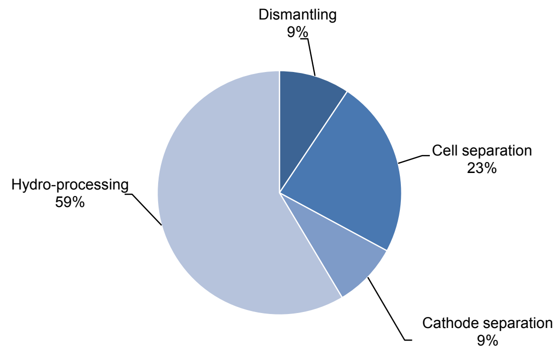
Exhibit 20: Unit CO2 emission for battery manufacturing by using virgin material versus recycling matl - CO2 emission can cut by 7-17% in most cases for recycling



Romare is a study by IVL Swedish Environmental Research Institute, authored by Mia Romare, published in 2017

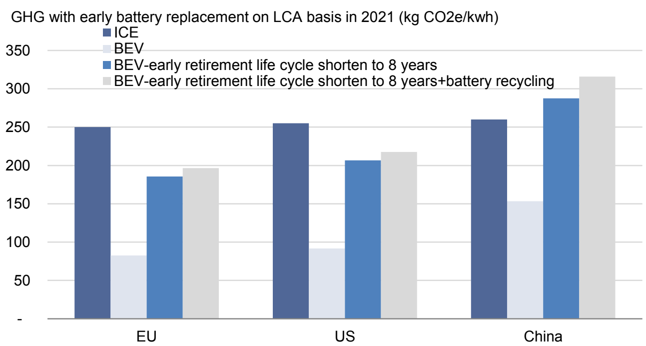
Source: Romare (2017): The Life Cycle Energy Consumption and Greenhouse Gas Emissions from Lithium-Ion Batteries, ICCT, Goldman Sachs Global Investment Research

Exhibit 21: Emission from recycling process - Hydrometallurgy



Source: Romare (2017): The Life Cycle Energy Consumption and Greenhouse Gas Emissions from Lithium-Ion Batteries, Goldman Sachs Global Investment Research

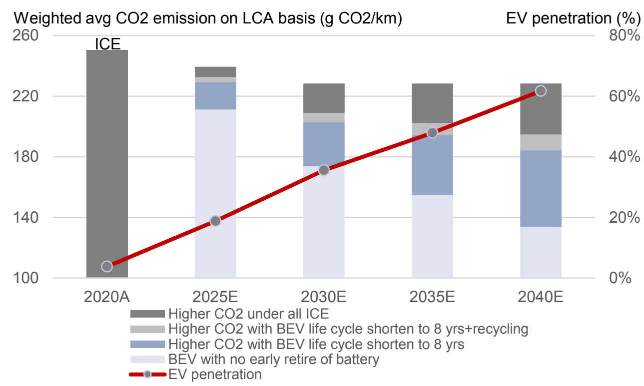
Exhibit 22: GHG emission on LCA basis for BEV, BEV with early battery retirement versus ICE



Battery replacement by using recycled materials under hydrometallurgy

Source: ICCT, Goldman Sachs Global Investment Research

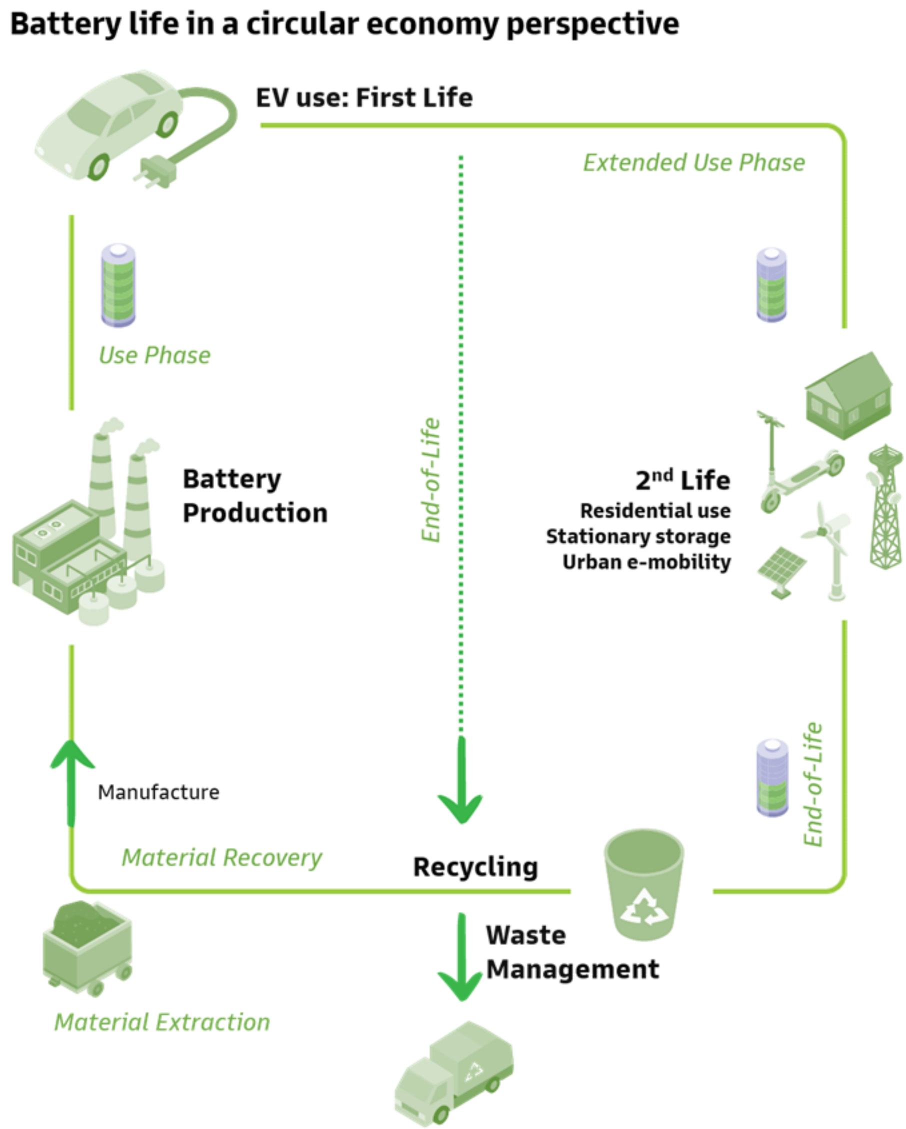
Exhibit 23: CO2 emission on LCA basis under rising EV mix and scenarios with battery early retirement and recycling



Source: ICCT, Goldman Sachs Global Investment Research

Path to the full recycling

Exhibit 24: EV battery life in a circular economy perspective



Source: Goldman Sachs Global Investment Research

Retired EV batteries' full collection and recycling are the ultimate targets for sustainable and eco-friendly EV development considering many valuable metals that are left and the big negative environmental impact if disposed improperly. To deliver the target, we believe stricter regulations are required to ensure the proper retirement, regulated collection and recycling treatment to fully utilise the retired batteries. On the other hand, EV battery recycling technology is still at early stages especially for pre-treatment processes including testing, dismantling, discharging and separation. Standardized EV battery is needed to improve the efficiency of pre-treatment under rising recycling volume.

Low collection rate in China - stricter regulations needed

EV battery recycling is still at early stages in China with regulation loopholes that lead to low collection rate for licensed recycling companies. By comparing to the systems that developed countries have set up for collecting retired batteries, we believe stricter regulation or potential law enforcement is needed to eliminate improper channel's participation and ensure retired EV batteries are fully utilised. Besides, appropriate consumer incentive system needs to be set up to encourage the timely retirement of EV batteries and collection by government authorized parties. Other than these, there is a lack of clear targets for EV battery recycling or collection in China compared to current EU/US proposed policies.

Retired EV battery collection/treatment rate is currently below 50% in China based on our estimate. However, this does not mean a lot of retired batteries are improperly disposed without treatment but more likely some retired batteries are collected by improper channels like small workshops or traders, or the delay in retiring due to lack of incentive/proper channels. Considering the environmental and safety risk associated with EV battery recycling and valuable critical materials left post retiring, Chinese government departments started to publish policies from 2016 to regulate EV battery recycling market including publishing MIIT (Ministry of Industry and Information Technology) whitelist of companies for EV battery recycling starting from 2018. As of now there're nearly 50 companies on MIIT published whitelist including the recent announced 3rd batch. However, the MIIT whitelist companies are not the only companies legally allowed to treat all retired EV batteries and these are more of a recommended list. According to SMM, the newly registered companies with major businesses in EV battery recycling in China have reached 15,000 in 2020, increased by nearly 3x yoy. This suggests a lot of retired EV batteries have gone to companies that are not in the MIIT whitelist and potentially without safety and environment licenses.

According to GGII, recycling whitelist companies in China have a total of 1.2mnt recycling capacity for retired EV battery, implying less than 10% utilisation rate and loss making for most licensed recycling companies. *Why is it hard for whitelist companies to collect retired EV batteries as much as they can?* The unlicensed workshops have more leverage to bid up battery acquisition price to get more retired EV batteries given their low treatment cost. With simple physical separation, they can sell for second-life reuse in low speed vehicles, which might lead to accidents. Or through simple pre-treatment to extract and sell part of metals left in retired EV batteries, small workshops can dispose other heavy metal contained materials without proper environmental treatment, leading to soil or water pollution and waste of valuable resources. These workshops mostly lack proper environment and safety treatment facilities and their storage and transportation also do not meet regulation required standards. Considering the future trend of EV prevailing over ICE, valuable and critical materials left post retiring, and safety and environmental risks for improper disposal of retired EV batteries, we believe stricter regulations or law enforcement to authorize licensed companies to treat all retired EV batteries need to be implemented. Under stricter regulations, we believe the improper disposal and treatment of retired EV batteries by unlicensed small workshops can be eliminated, which currently pose safety and environmental risk, and with low efficiency of recycling in terms of metal extraction.

MIIT whitelist for retired EV battery recycling companies

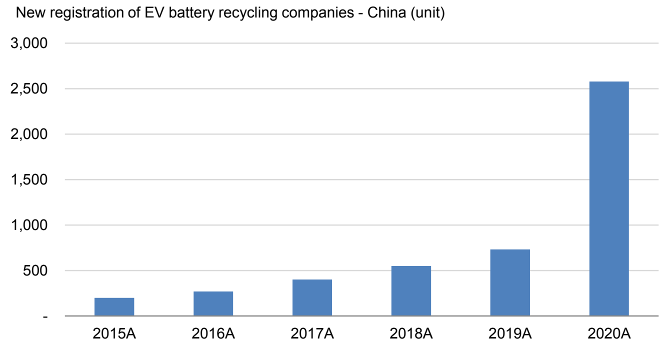
MIIT published the first batch of whitelist companies in 2018 and only five companies were on the list - Quzhou Huayou, Guangzhou Haopeng, Jingmen GEM, Hunan Bangpu and Guangdong Guanghua Technology. In early 2021, MIIT published the second batch of whitelist companies with 22 companies on the list, including 9 companies with recycling license and others with second life reuse license. All the whitelist companies have met the retired EV battery recycling industry standards published by MIIT. MIIT recently announced the third batch of whitelist companies with 20 companies on the list, including 11 companies with recycling license and others with second life reuse license.

However, it’s not compulsory to use whitelist companies to process retired batteries under the current regulation, but can transfer legal risk of potential accidents during retired EV battery treatment process or reuse to whitelist companies.

We believe retired EV battery recycling will be a highly regulated market in the future and only licensed players would be able to participate in the recycling process. We expect proper collection and treatment of retired EV battery would become top priority of governments considering the exponential rise in the EV market size (13x of current level by 2040E, per GSe). We saw an acceleration in regulations published on retired EV battery recycling in China since 2015. Although these policies are mostly focusing on regulating licensed companies and lack of law enforcement or penalty on unregulated behaviors, these policies lay the foundation for potential law enforcement in some areas to help complete EV battery self-circulation. The most important one being EV battery tracking system introduced by MIIT in 2018, which we believe can improve regulated battery recycling in the future. Specifically, in July 2018, MIIT published the Provisional Regulation of EV Battery Recycling and Utilisation Tracking System. The regulation requires the registration of each new EV car to help track EV batteries by allocating a unique number. Under this system, information is collected on battery production, sales, usage, battery retirement from EVs, recycling and second life. Responsible parties involved in the process will be monitored to meet the proper usage/treatment of EV battery. According to China EV Battery Association, battery and car manufacturers that have registered to the system have increased to current 80-90% from 30-40% in 2018 when first introduced. The tracking system can help make the EV batteries stay within the EV supply chain loop. However, we believe law enforcement is needed to make sure all these regulation are strictly followed. The industry participants have pushed for law enforcement of retired EV battery recycling since 2018 and MIIT is also pushing for accelerating the law enforcement of retired EV battery recycling since 2020.

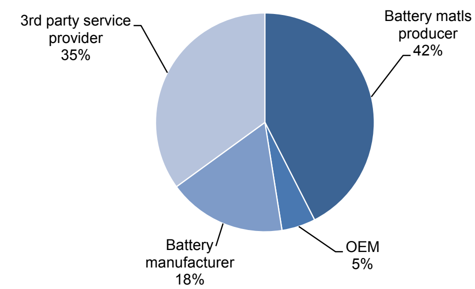
There is also a lack of clear targets for EV battery recycling or collection in China. In EU’s proposal of batteries and waste batteries regulation, it lays out quantitative targets for recycling efficiencies, recovery rates and minimum recycled material content in batteries. The US also has set recycling targets for lithium batteries by 2030.

Exhibit 25: New registration of EV battery recycling companies - China



Source: SMM, Data compiled by Goldman Sachs Global Investment Research

Exhibit 26: Whitelist company breakdown by type - China



Source: MIIT, Data compiled by Goldman Sachs Global Investment Research

Exhibit 27: US/EU policy related to EV battery recycling

EU proposal for new Battery Regulation related to EV battery recycling	
near term targets	**long term targets**
Recycling efficiency of waste batteries at no later than 1st Jan 2025: 65% for lithium-ion batteries (preferred option)	Recycling efficiency of waste batteries at no later than 1st Jan 2030: 70% for lithium-ion batteries
Material recovery rates by 1st Jan 2026 (preferred option): Cobalt-90%, Nickel-90%, Lithium-35%, Copper-90%	Material recovery rates by 1st Jan 2030: Cobalt-95%, Nickel-95%, Lithium-70%, Copper-95%
Minimium recycled material content in batteries from 1st Jan 2030: cobalt-12%, lead-85%, lithium-4%, nickel-4%	Minimium recycled material content in batteries from 1st Jan 2035: cobalt-20%, lead-85%, lithium-10%, nickel-12%
Mandatory carbon foodprint declaration (preferred option)	Carbon footprint performance classes and maximum carbon thresholds for batteries as a condition for placement on the market
Mandatory declaration of levels of recycled content, in 2025	Mandatory declaration of levels of recycled content, in 2030 and 2035
US National Blueprint For Lithium Batteries on EV battery recycling	
near term targets - 2025	**near term targets - 2030**
Establish federal recycling policies to promote collection, reuse, and recycling of lithium-ion batteries	Create incentives for achieving 90% recycling of consumer electronics, EV, and grid-storage batteries
	Develop federal policy requiring the use of recycled materials in cell manufacturing materials streams

Source: European Commission, FCAB, Data compiled by Goldman Sachs Global Investment Research

Exhibit 28: China policies on EV battery recycling

EV battery recycling policies - China		
2016	State Council	Set up EV battery recycling system. EV OEMs and battery manufacturers should be responsible for setting up retired EV battery recycling network. Set up tracking system for battery life usage. Set up EV battery recycling pilots project in Shenzhen and expand to nationwide. 建立电动汽车动力电池回收利用体系。电动汽车及动力电池生产企业应负责建立废旧电池回收网络，利用售后网络回收废旧电池。动力电池生产企业应实行产品编码，建立全生命周期追溯系统。率先在深圳等城市开展电动汽车动力电池回收利用体系建设，并在全国逐步推广。
2016	MEE	Publish standards for EV battery recycling process including collection, storage, transportation and recycling to control the environmental risks during recycling process. 发布废电池污染防治技术政策。对锂离子电池的回收、运输、贮存、再生提出明确规定，以期控制锂离子电池回收利用过程中的环境风险。
2018	MIIT, MEE, Energy Bureau	Introduction of Extended Producer Responsibility (EPR) system for EV retired battery recycling, and set OEM as responsible party to collect retired EV batteries. 实行生产者责任延伸制度，汽车生产企业承担动力蓄电池回收的主体责任，相关企业在动力蓄电池回收利用履行相应责任， 保障动力蓄电池的有效利用和环保处置。
2018	MIIT, MEE, Energy Bureau	Publish the Provisional Regulation of EV Battery Recycling and Utilisation Tracking System. The regulation requires the registration of each new EV car to help track EV batteries by allocating a unique number. Under this system, information is collected through battery production, sales, usage,retire, recycling and second life. 电池生产企业应与汽车生产企业协同，按照国家标准要求对所生产动力蓄电池进行编码，汽车生产企业应记录新能源汽车及其动力蓄电池编码对应信息。电池生产企业、汽车生产企业应及时通过溯源信息系统上传动力蓄电池编码及新能源汽车相关信息。
2018	MIIT	Publish of the first batch of whitelist companies that can meet EV battery recycling standards 发布符合《新能源汽车废旧动力蓄电池综合利用行业规范条件》企业名单（第一批）
2018	NTCAS(National Technical Committee of Auto Standardization)	Publish requirement for recycling materials of automobile power battery. Require lithium-ion power batteries recycling yield should be no less than 98% for nickel, cobalt, and manganese, lithium no less than 85%. 发布车用动力电池回收利用材料回收要求。锂离子动力电池材料中镍、钴、锰元素的综合回收率应不低于98%，锂元素的回收率应不低于85%。

Source: MIIT, MEE, NDRC, Data compiled by Goldman Sachs Global Investment Research

How developed countries have built battery recycling systems

China lags behind developed countries in building proper battery recycling systems. In traditional lead-acid battery, NDRC targets to reach collection rate above 70% by 2025E, versus close to 100% in developed countries. Compared to developed countries, China is lacking on:

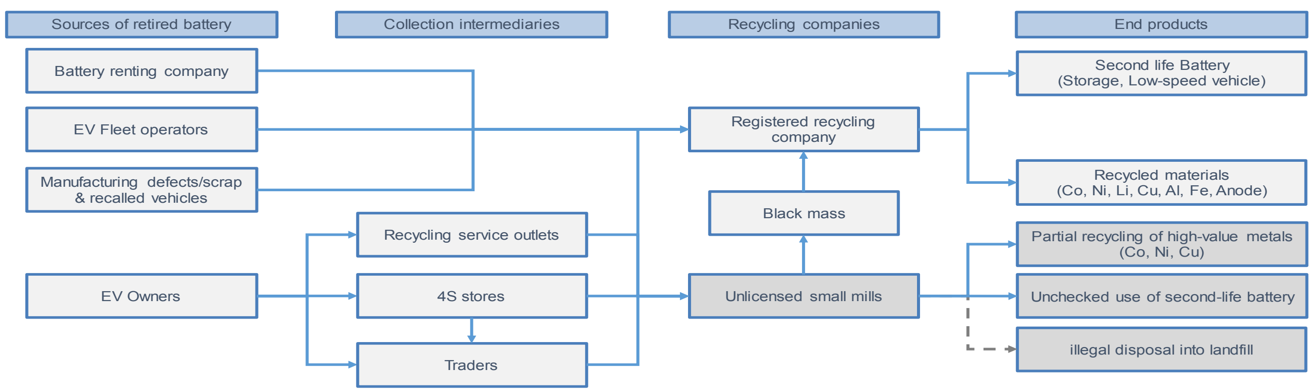
- Enforcement mechanism, collection and recycling targets: compared to US/Japan/EU, battery recycling scheme in China is not legally binding and does not have an effective enforcement mechanism. For example, EU has proposed a new regulatory framework for batteries collection and recycling, with 1) minimum collection target for both portable battery, automotive, and industrial batteries, 2) mandatory carbon footprint declaration for industrial and EV batteries, and maximum carbon threshold requirement for batteries to be sold, 3) mandatory levels of recycled content in industrial, EV and automotive batteries by 2030-2035. Japan has also laid out specific mechanism to enforce the recycling scheme. When the recycling efforts by a producer do not meet requirements set by the government, the producer could potentially be asked to take necessary measures and a penalty could be imposed.

- Strict regulation on market participants: a large proportion of the retired battery in China went to unlicensed small mills, where batteries are usually not properly treated and waste is not properly disposed. In comparison, there is regulatory requirement in Europe to ensure retired batteries are delivered to an approved battery treatment operator or exporter for treatment and recycling. Similarly, retired batteries in US must be delivered and recycled at a universal waste handler or an authorized recycling facility.
- Consumer incentive system set up: lack of consumer awareness and incentive also partly contributes to the relatively low collection rate in China. In US, consumers are required to pay a battery core charge at purchase in over 30 states, which will be refunded when the battery is returned. The core charge system provides additional consumer incentive to promote battery recycling.

Recycling technology is still evolving with EV battery technology

Despite EV battery recycling still at a very early stage, metal extraction technology from waste electronics, alloy and consumer batteries have developed for years, and has laid the foundation of high recovery rate of key minor metals including nickel, cobalt and lithium under hydrometallurgical method in recent years. Key technology gaps for EV battery recycling are mostly in retired EV battery pre-treatment stage including lack of more efficient, large-scale battery dismantling, discharging and testing due to current inconsistent standards for EV battery production. Technology development in metal extraction are mostly on improving recovery rate, shortening the recycling process and lowering the cost.

Exhibit 29: Simplified EV battery recycling process



Source: Goldman Sachs Global Investment Research

Retired EV battery recycling 101

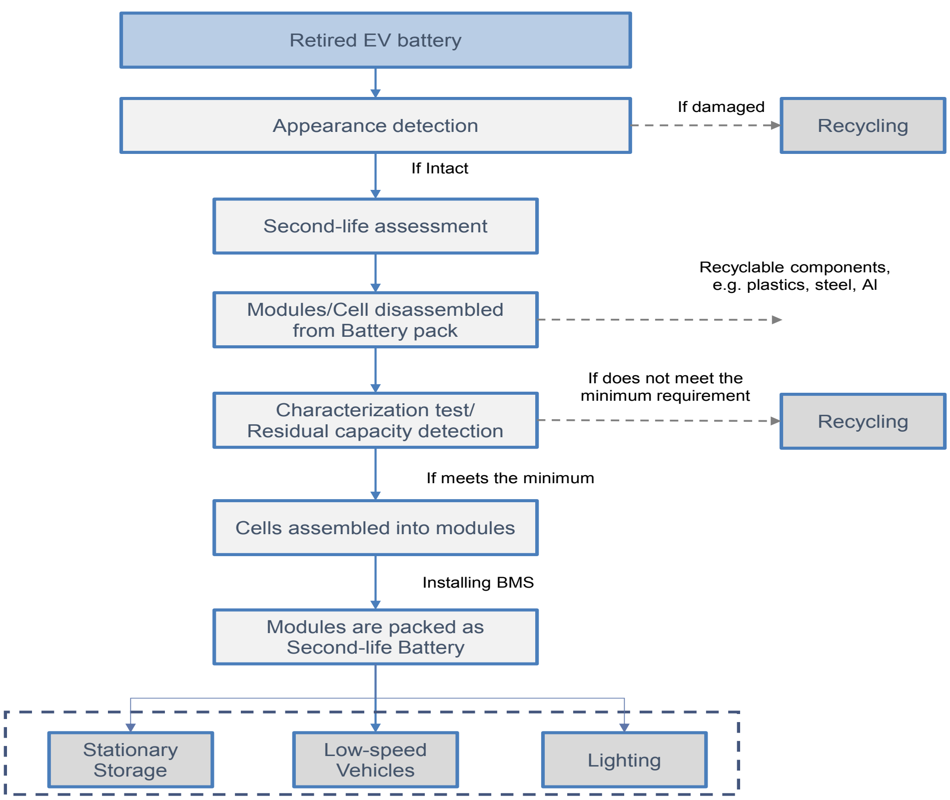
EV batteries typically need to retire from EVs when the capacity falls below 80% of the initial level. Retired EV batteries are either recycled to recover raw materials or repurposed for second life applications. The entire recycling process includes the following steps:

- **Collection & Transportation:** Major sources of retired EV batteries include EV owners, fleet operators or bus operators, battery renting companies, and manufacturing scraps. These batteries are normally collected by professional recycling companies, traders, and 4S stores, and then shipped to established recycling facilities or unlicensed small mills for further treatment. In China, retired batteries are subject to special storage and transportation requirements for Dangerous Goods due to its high flammable and explosive risks.
- **Second-life reuse:** batteries with 40-80% of their initial capacity could potentially be repurposed for second-life application, e.g. energy storage and low-speed vehicles. Modules and cells will first be disassembled from the battery pack. The disassembled cells will then go through a series of characterization tests to determine the residual capacity. Cells that meet the minimum requirements for second-life reuse will be re-assembled into modules and battery pack, with the installation of a new battery management system (BMS). Second-life reuse has higher requirements for retired EV battery in terms of consistency in capacity, size,

inner structure, economics versus new battery.

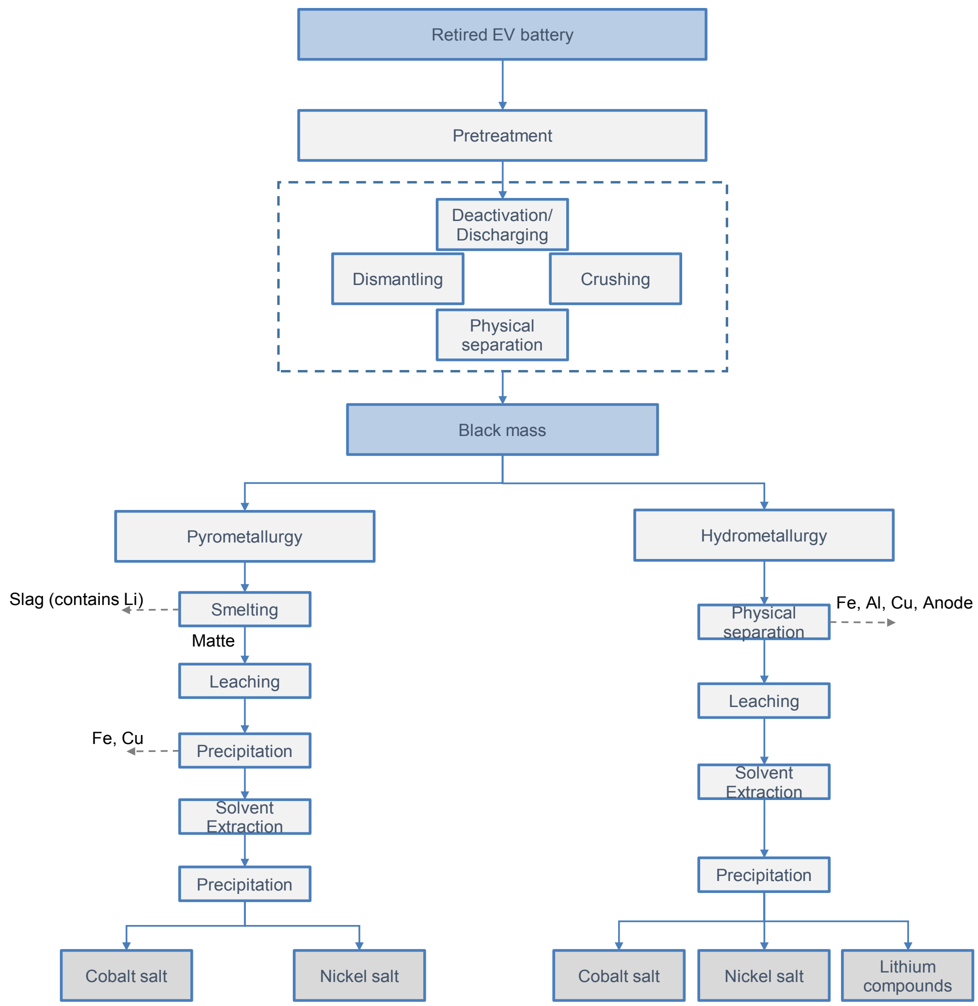
- **Recycling:** batteries that do not meet the minimum requirements for second-life reuse are recycled. Retired batteries need to be pre-treated first, including discharging, dismantling, crushing, and physical separation. Fine powder called "black mass" is produced after the pretreatment. Currently, below are the major processes for recycling treatment but only pyrometallurgical and hydrometallurgical are operating at industrial scale:
  - **Pyrometallurgical process,** batteries are first burned in a smelter, where electrolyte and plastics are burned away and graphite anode and aluminum are oxidized in a reduction reaction with metals. The end products include a metal alloy, which contains cobalt, copper, and nickel, and slag. Slag contains lithium and aluminum oxide. The alloy and slag can be further leached with acid to separate and recover pure materials.
  - **Hydrometallurgical process**, black mass is dissolved in strong acid such as sulphuric acid to break up the crystal structure. Various metal salts including cobalt sulphate and nickel sulphate can be recovered through solvent extraction and precipitation. Hydrometallurgical process is the dominant recycling method in China.
  - **Direct recycling:** direct recycling will enable recovery of higher value cathode materials in a condition suitable for direct re-entry into EV battery production, providing a lower-cost method for battery recycling.

Exhibit 30: EV Battery second-life reuse process



Source: Warner (2015)-The Handbook of Lithium-Ion Battery Pack Design, Data compiled by Goldman Sachs Global Investment Research

Exhibit 31: EV battery metal recycling processes - Pyrometallurgy vs Hydrometallurgy



Source: Argonne National Laboratory, Data compiled by Goldman Sachs Global Investment Research

More technology gaps in recycling pre-treatment stages

With rapid development in battery technology and rising volume of retired EV battery, further technological improvements are required to provide more cost-saving and efficient solution of battery recycling and second-life reuse. Under current technology, metal extraction from black mass (fine powders post physical separation and dismantling of retired EV batteries) can reach a very high level of recovery rate or efficiency rate under hydrometallurgy method with cobalt and nickel recovery rate above 95% and lithium above 90%, but technology is still at early stages for pre-treatment processes including testing, dismantling, discharging and separation due to continued battery technology development and inconsistency in battery system standards. We highlight the following technology gaps and the R&D efforts:

Battery dismantling: Currently, battery dismantling is mostly done manually, leading to an inefficient separation process, high labour cost, and rising safety risks. Automated battery disassembly would be increasingly important under rising recycling scale. However, the automation of EV battery disassembly technology remains highly challenging due to lack of standardization in battery design and changing battery types over time. According to Harper (2019): "Recycling lithium-ion batteries from electric vehicles", initial efforts to develop automated disassembly include establishing labeling standards for EV batteries and leveraging AI capabilities for handling diverse waste materials. With trends of standardization of battery system, automation of disassembly will decrease the cost and reduce the safety risk. Currently, most of Chinese recycling companies use the process combining manual and automated disassembly process to increase efficiency.

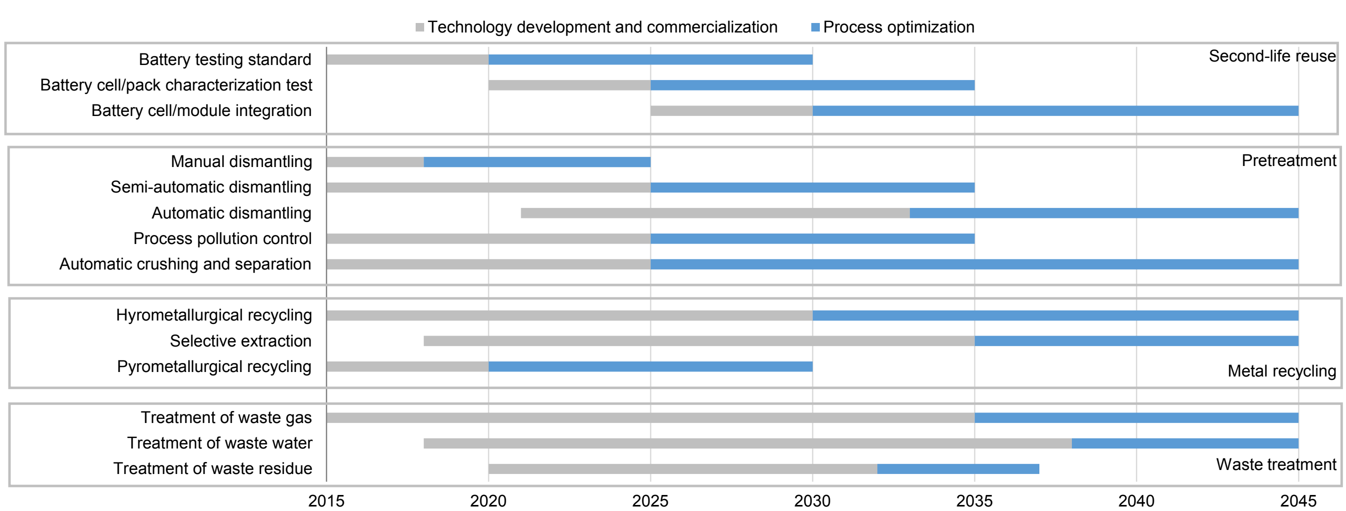
Battery discharging: the retired EV batteries need to be discharged before physical separation of battery to lower the danger from chemical reaction in batteries, who still have power left inside. For the safe disassembly of retired batteries, a complete discharge of the energy store in battery is preferable. Currently the common methods for discharging include 1) water or brine immersion, 2) freezing, 3) thermal treatment with water immersion the most commonly used one in China. However, the key technology bottleneck need to be solved for discharging before larger scale of recycling is faster and more efficient discharging with safety. Although the water immersion method is simple, the discharge process needs at least 24 hours or more with respect to the voltage and temperature, limiting scale for recycling. The hydrogen released from the anode during discharging and chloride ion in the solution will corrode the electrode and contaminate electrolyte system, leading to safety issues for the working environment. On the other hand, the battery casing could likely be corroded and cause part of metal in cathode to dissolve/leak into the discharge solution, leading to lower recovery rate of metals.

Second-life battery: the major hurdles to second-life repurposing include 1) the wide varieties of battery designs (types, chemistry, form factors) complicates the assessment and matching process and makes second-life battery more costly; 2) Inaccurate and inconsistent measurement of cells' state of health posed additional risks for second-life reuse, such as shorter battery life and higher safety risk. The presence of technical gap is also reflected in China's plan to ban large energy storage project from using repurposed EV battery following a fire incident at an energy storage power station in Beijing. 3) loss of cost advantage with declining new battery production cost trend. We estimate a second-life LFP ESS battery cost is currently around Rmb0.64-0.68/Wh, with the main costs being battery acquisition (56%), testing and packaging (9%), and labour (13%), versus new ESS battery of around Rmb1.20/Wh. Assuming 80% residual capacity, lifespan of second-life battery in energy storage is around 10 years, 6-7 years shorter than new battery. This implies an annualized battery cost of Rmb0.11/Wh for second-life battery, 29% lower than that of new battery. However, potential challenge for the second-life application comes from the expected decline in new battery production cost, which will lead to a continuous cost convergence between new battery and second-life battery.

Technological improvement could help lower the cost of battery repurposing, for example, battery management systems could be used to record state of health information of cell and modules, thus accelerating the battery assessment process. Increased collaboration among OEMs and battery producers and increased standardization in battery design would also enhance the safety and economics of battery repurposing.

Direct recycling aims to recover active materials such as NMC cathode while preserving the compound structure and functionality, thus maximizing the economic value and simplifying treatment processes. According to ReCell Center, direct recycling has the potential to lower ternary cathode production cost by 40-50% compared to virgin materials. In addition, direct recycling could enhance the economic feasibility of LFP battery recycling if LiFePO4 can be recovered. Although direct recycling could be a promising development, it is still a lab-scale technology with significant amount of technical hurdles to overcome, including an effective method to separate cathode from other materials (anode, Cu foil, and aluminum foil) and a relithiation process that prepares the recovered materials for EV battery application.

Exhibit 32: Recycling Technology roadmap - China



Source: ATCRR, Chinese Academy of Sciences, Data compiled by Goldman Sachs Global Investment Research

China vs Global on recycling technologies

With our expectation of accelerating EV battery retirements in the future, both China and developed countries including US and Europe have been expanding EV battery recycling facilities and increase R&D in recycling technology. Compared to the counterparts in developed countries, EV battery recycling facilities in China have a different technology mix and larger scale, and lower unit processing cost:

- Among the industrial scale recycling projects (>1,000tpa), hydrometallurgical process is the dominant technology in China, while pyrometallurgical process is mostly used in US and Europe. While both technologies are mature and commercialized, hydrometallurgical process has the advantage of lower carbon intensity and higher recovery rate of most battery materials. Since traditional pyrometallurgical process requires a relatively high cobalt content for economics, a few specialized recycling companies in EU such as Umicore are combining pyrometallurgical process with hydrometallurgical processes to increase recovery rate and further extract more metals.
- Due to an early start in EV battery retirement - we estimate China accounts for 50-70% of global EV battery recycling by 2030, recycling facilities in China currently are operating at a larger scale than those in developed markets. For example, GEM and Huayou Cobalt both have EV battery recycling capacity of over 60ktpa, versus 20ktpa for Eramet, 7ktpa for Umicore.
- Benefiting from the economies of scale, unit battery processing cost in China is around US$1,300/t, much lower than US$1,500-5,400/t in developed markets. Higher cost in developed markets is mainly driven by higher transport and dismantling cost, as well as higher cost from the metallurgical processes due to lower battery throughput.

Despite the late start, US and EU are ramping up recycling scale and accelerating the development of new technology including direct recycling and automated disassembly. For example, US department of Energy (DoE) has launched ReCell as a national advanced battery recycling R&D center, focusing on developing direct recycling process and enhancing recovery of other raw materials including electrolyte and anode. In addition, researchers at DoE’s Oak Ridge National Laboratory have developed an automated EV battery disassembly system that can be adopted for different types of EV batteries. In comparison, more R&D efforts and coordination at national level remain to be seen in China despite MIIT announcement to push forward the development of key recycling technologies such as automated disassembly and active materials separation.

Competition to drive return to a normalized level

In the future, we believe OEMs and battery manufacturers can gain more bargaining/pricing power over raw material resources with rise of EV battery metal recycling. Currently under tight metal supply including lithium, nickel and cobalt with strong EV growth, OEMs and battery manufacturers suffer from margin contraction due to metal price surge and are unable to pass through to downstream customers. OEMs and battery manufacturers have very low bargaining power in terms of raw material price hikes as most of the mining resources are controlled by several professional mining companies. However, with EV battery recycling starting to take off, most of retired EV batteries will be controlled by OEMs under EPR system and also with their 4S stores as last mile connections to EV customers while battery scraps are mostly controlled by battery manufacturers. Most of the recycling companies can only serve as service providers as they need to cooperate with OEMs and battery manufacturers to procure retired EV batteries. With recycled metals including lithium, nickel and cobalt supplying 70-80% of battery demand in the long run, we believe mining supply for these metals will not be the bottleneck for EV development in the future with alternative raw material source from EV battery metal recycling. And under a service provider business model, we believe return for EV battery metal recycling should gradually come down to normal level at 10% EBTIDA/capex in the long run versus current 60-90% return for battery material producers or 0-19% for 3rd party service providers under full utilisation of capacity. For battery material producers, rising volume from EV battery metal recycling provides alternative raw material sources, who are currently relying on upstream mining resources to grow its volume.

Stricter regulation will help increase market share for licensed recycling companies

Current EV battery recycling market is fragmented with various players including OEMs, battery manufacturers, battery material producers, battery renting companies, professional third-party service providers, grid operators, EV fleet operators, traders and unlicensed family workshops. Comparing to consumer electronic batteries, dissemble of EV batteries from EVs is more difficult and need professionals to complete the replacement. EV car owners need to go to 4S stores or EV battery renting companies to replace the battery. Under China's Extended Producer Responsibility (EPR) system for EV retired battery recycling introduced in 2018, OEMs are set as responsible parties for collecting retired batteries through recycling services outlets and 4S stores. As OEMs lack the treatment facilities and licenses, they cooperate with battery manufacturers, battery material producers, professional third party recycling companies to recycle collected retired batteries. In the future, we believe most of retired EV batteries will be controlled by OEMs under EPR system and also with their 4S stores as last mile connections to EV car customers while battery scraps are mostly controlled by battery manufacturers. The recycling companies including battery material producers and professional third-party service providers need to cooperate with OEMs and battery manufacturers by providing "one-stop service" including testing, storage, transportation, second life reuse or recycling pending on license they own to get retired batteries or battery scrap.

There're currently two major types of participants in retired EV battery recycling:

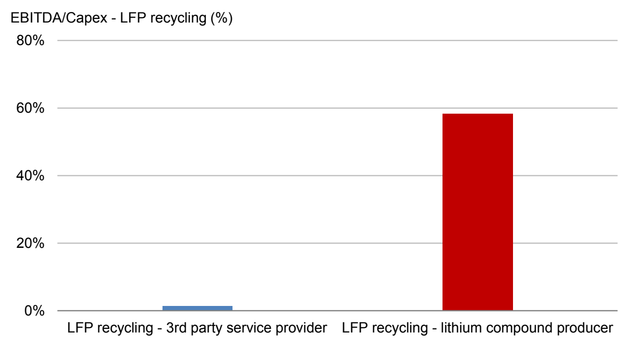
- Battery material producers: those who are mostly engaging in battery material production and existing suppliers for battery manufacturers and OEMs. These companies expanded to EV battery recycling to use recycled raw materials mixing with mining raw materials to produce battery materials. Key players include GEM, Huayou, Ganfeng, Bangpu and Umicore.
- 3rd party service providers: those who are only engaging in recycling business and sell recovered battery materials. Key players include Hengchuang Ruineng, Beijing Saidemei and Li-cycle.

Based on MIIT's announced whitelist company types, professional recycling companies account for the majority including 42% of battery material producers and 35% of 3rd party service providers while battery manufacturers and OEMs only account for 23% of the announced whitelist companies. Therefore, professional recycling companies are likely to benefit more from EV battery recycling acceleration under more regulated market. Furthermore, we believe battery material producers are better positioned comparing to 3rd party service providers for below aspects:

- Entry barrier in EV supply chain: as existing suppliers for battery manufacturers and OEMs, battery material producers already passed the product qualification in EV supply chain while 3rd party service provider still needs time for qualification to enter the EV supply chain.
- More high-end products: battery material producers are more integrated in terms of EV battery recycling business as they not only have recycling capacity but also have back-end compound capacity to produce more high-end products from recovered metals. For example, Ganfeng can sell lithium carbonate or hydroxide rather than lithium chloride and GEM/Huayou can sell precursor or cathode rather than nickel sulphate and cobalt sulphate from recovered metals through battery recycling.
- Higher return: as battery material producers are more integrated with existing compound or sulphate capacity, there's no need for extra investment in back-end capacity to sell high-end products. We estimate battery material producers can get 60-90% return (EBITDA/capex) for LFP or ternary battery recycling while 3rd party can only get 0-19% return under full utilisation of capacity. Noticeably, 3rd party service providers can still get nearly 20% return for recycling retired ternary battery.
- Closer relationship with OEMs and battery manufacturers: as existing suppliers, battery material producers already set up strong relationship with customers including OEMs and battery manufacturers. Hence, they can build a strong closed-loop supply chain that once OEMs and battery manufacturers collected retired battery, they can send to battery material producers who have recycling capacity to produce battery materials and circulate back to OEM and battery manufacturers.

Overall, with growing size of retired EV battery market, we expect competition to increase under current high return especially in the ternary battery market. As most of the recycling companies don't have control over retired EV battery or battery scrap resources, the business model is likely to serve as recycling service provider for integrated recycling companies including battery material producers. As a result, we believe the coalition with OEMs or battery manufacturers are the most important to secure retired battery resources. We note Bangpu under CATL who are currently recycling CATL's battery scraps or retired batteries to produce battery materials, GEM signed MOU with EVE Battery to provide recycled nickel products through retired battery and battery scraps from EVE Battery, Li-Cycle signed offtake agreement with LG Chem and LG Energy Solution on a closed loop arrangement to recycle battery scraps or other EV battery materials and supply LG Chem and LG Energy Solution battery manufacturing plants with recycled nickel battery-grade product.

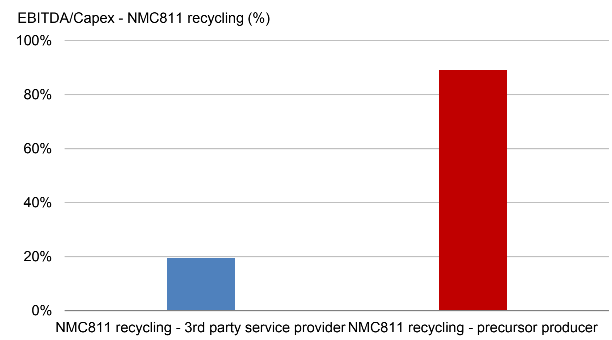
Exhibit 33: EBITDA/Capex of LFP recycling - 3rd party vs battery matl producer



Based on 2021 average price

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 34: EBITDA/Capex of NCM811 recycling - 3rd party vs battery matl producer



Based on 2021 average price

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 35: Valuation comps - Global cathode and recycling peers

Company	Ticker	Rating	Target	Price	ccy	Mkt cap	PE			PB			ROE			EV/EBITDA			Div yield		
Price as of 10-Jan-2022				Local		US$ mn	20A	21E	22E	20A	21E	22E	20A	21E	22E	20A	21E	22E	20A	21E	22E
GEM	002340.SZ	Buy	16.00	9.82	CNY	7,373	106.3	45.8	19.9	3.3	3.3	2.9	3%	7%	14%	15.8	19.5	11.4	0.3%	0.4%	0.9%
Huayou Cobalt	603799.SS	Buy	128.00	98.01	CNY	18,158	93.4	34.2	22.0	11.0	6.1	5.0	12%	18%	23%	21.5	22.9	13.0	0.5%	0.6%	0.9%
Zhongwei	300919.SZ	NC	NA	134.31	CNY	12,768	163.8	72.9	41.5	19.9	13.6	10.4	18%	21%	27%	n.a.	55.4	35.7	0.1%	0.2%	0.4%
Xiamen Tungsten	600549.SS	NC	NA	21.87	CNY	4,865	50.0	22.8	17.5	4.0	3.6	3.1	8%	16%	19%	n.a.	n.a.	n.a.	0.7%	2.0%	3.0%
Ronbay	688005.SS	NC	NA	126.51	CNY	8,894	263.6	67.3	33.4	12.5	10.6	8.0	5%	16%	25%	n.a.	52.6	26.7	0.1%	0.2%	0.6%
Eastspring	300073.SZ	NC	NA	79.79	CNY	6,341	90.6	40.4	32.2	8.7	7.3	6.2	12%	19%	20%	68.4	35.0	25.5	0.2%	0.4%	0.5%
Shanshan	600884.SS	NC	NA	28.65	CNY	9,643	286.5	17.6	17.8	3.6	3.2	2.8	1%	18%	16%	55.7	15.9	13.2	0.3%	1.1%	1.1%
China peers							**150.6**	**43.0**	**26.3**	**9.0**	**6.8**	**5.5**	**8%**	**16%**	**20%**	**40.4**	**33.6**	**20.9**	**0.3%**	**0.7%**	**1.1%**
UMICORE	UMI.BR	Neutral	37.00	35.60	EUR	9,695	28.9	12.6	14.4	3.3	2.8	2.5	12%	22%	17%	13.5	7.7	8.8	1.9%	2.2%	2.5%
ECOPRO	086520.KS	NC	NA	98,300	KRW	2,004	77.1	7.2	28.8	5.7	3.9	3.6	8%	40%	16%	20.8	9.2	6.0	0.4%	0.4%	0.4%
Li Cycle	LICY	NC	NA	8.69	USD	1,418	n.a.	n.a.	n.a.	n.a.	2.7	2.5	n.a.	-10%	-4%	n.a.	n.a.	n.a.	0.0%	0.0%	n.a.
SUMITOMO CHEMICAL	4005.T	NC	NA	570	JPY	8,194	20.2	7.4	7.5	0.9	0.8	0.8	5%	12%	11%	7.0	5.2	5.0	2.6%	3.8%	3.9%
POSCO CHEMICAL	003670.KS	NC	NA	130,500	KRW	8,430	269.1	69.8	61.1	n.a.	4.4	4.2	n.a.	9%	7%	85.3	42.2	33.7	0.2%	0.3%	0.3%
Global peers							**99**	**24**	**28**	**3.3**	**2.9**	**2.7**	**8%**	**15%**	**9%**	**31.7**	**16.1**	**13.4**	**1.0%**	**1.3%**	**1.8%**

Source: Company data, Thomson Reuters, Goldman Sachs Global Investment Research

Exhibit 36: Valuation comps - Global and China lithium peers

Company	Ticker	Rating	Target	Price	ccy	Mkt cap	PE			PB			ROE			EV/EBITDA			Div yield		
Price as of 10-Jan-2022				Local		US$ mn	20A	21E	22E	20A	21E	22E	20A	21E	22E	20A	21E	22E	20A	21E	22E
Ganfeng Lithium	1772.HK	Buy*	246.00	115.90	HKD	19,392	120.5	47.0	26.4	11.9	7.9	6.4	10%	17%	24%	31.5	34.9	19.3	0.9%	0.8%	1.5%
Ganfeng Lithium	002460.SZ	Buy	256.00	129.06	CNY	26,420	164.2	64.0	36.0	16.2	10.8	8.7	10%	17%	24%	48.2	47.0	25.8	0.5%	0.6%	1.1%
Tianqi Lithium	002466.SZ	NC	NA	95.48	CNY	22,127	n.a.	n.a.	54.8	26.9	15.9	11.8	13%	26%	23%	44.4	22.2	19.9	0.1%	0.3%	0.3%
Sichuan Yahua	002497.SZ	NC	NA	25.93	CNY	4,691	78.6	32.2	20.9	5.7	4.7	3.8	11%	15%	19%	n.a.	n.a.	n.a.	0.1%	0.2%	0.2%
Youngy	002192.SZ	NC	NA	114.55	CNY	4,669	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0.0%	0.0%	n.a.
SZChengxin Lithium Gp	002240.SZ	NC	NA	50.63	CNY	6,880	n.a.	51.5	37.7	10.3	9.2	7.6	1%	18%	20%	n.a.	n.a.	n.a.	0.0%	0.6%	0.8%
Nanjing Hanrui	300618.SZ	NC	NA	76.57	CNY	3,722	66.0	30.9	25.8	6.0	5.0	4.3	12%	17%	17%	n.a.	n.a.	n.a.	0.3%	0.7%	0.8%
Zhejiang Huayou	603799.SS	Buy	128.00	98.01	CNY	18,158	93.4	34.2	22.0	11.0	6.1	5.0	12%	18%	23%	21.5	22.9	13.0	0.5%	0.6%	0.9%
Regional peers							**121.1**	**48.7**	**35.2**	**14.2**	**9.7**	**7.7**	**9%**	**18%**	**22%**	**41.3**	**34.7**	**21.7**	**0.3%**	**0.4%**	**0.8%**
Albemarle Corp.	ALB	Sell	205.00	226.97	USD	24,180	55.0	54.9	33.4	5.7	4.5	4.6	11%	9%	14%	15.2	35.2	25.1	1.8%	0.7%	0.7%
Livent Corp.	LTHM	Sell	23.00	23.48	USD	3,430	n.a.	148.2	44.1	6.2	5.8	5.3	-1%	4%	12%	70.1	57.3	26.6	0.0%	0.0%	0.0%
SQM	SQM	NC	NA	48.36	USD	13,813	76.8	28.8	15.9	n.a.	4.3	4.0	8%	18%	26%	26.3	14.6	9.2	1.1%	2.2%	3.9%
Lithium Americas	LAC.TO	NC	NA	33.91	CAD	3,215	n.a.	n.a.	n.a.	17.9	n.a.	n.a.	-20%	-14%	-1%	n.a.	n.a.	111.7	0.0%	0.0%	n.a.
Galaxy Resources	GXY.AX	NC	NA	5.28	AUD	0	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0.0%	0.0%	n.a.
Pilbara Minerals	PLS.AX	NC	NA	3.52	AUD	7,504	n.a.	23.9	17.1	17.8	9.8	6.1	-10%	50%	39%	n.a.	15.4	12.9	0.0%	0.0%	n.a.
Allkem Ltd	AKE.AX	NC	NA	10.88	AUD	4,967	n.a.	42.3	33.9	5.6	4.4	3.9	-12%	13%	14%	n.a.	n.a.	19.8	0.0%	0.0%	n.a.
Lithium Australia	LIT.AX	NC	NA	0.12	AUD	88	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0.0%	0.0%	n.a.
NEO Lithium	NLC.V	NC	NA	6.37	CAD	710	57.9	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0.0%	0.0%	n.a.
Bacanora Lithium	BCN.L	NC	NA	67.25	GBP	352	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0.0%	0.0%	n.a.
Mineral Resources	MIN.AX	Neutral	40.00	60.73	AUD	8,203	34.2	10.4	21.9	5.0	3.6	3.5	15%	34%	16%	3.6	3.3	10.5	6.3%	7.9%	2.3%
Lithium Power Intl	LPI.AX	NC	NA	0.58	AUD	145	n.a.	n.a.	n.a.	4.5	5.3	6.4	-16%	-22%	-31%	n.a.	n.a.	n.a.	0.0%	0.0%	n.a.
Average global peers							**56.0**	**51.4**	**27.7**	**9.0**	**5.4**	**4.8**	**-3%**	**12%**	**11%**	**28.8**	**25.2**	**30.8**	**0.8%**	**0.9%**	**1.7%**

Source: Company data, Thomson Reuters, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Joy Zhang, Trina Chen and Arthur Deng, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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